

DBMS Lab Sheet 4 – SQL User Management & Access Control

Name: Sakshi Garg

Roll No: 2401010178

Course: B.Tech CSE (AI & DS)

Subject: Database Management Systems Lab

Part A: User Creation

1 Create user1 with password:

1	•	CREATE database Abc;
2	•	use Abc;
3		
4	•	CREATE USER user1 IDENTIFIED BY 'password123';

	Time	Action	Response	Duration / Fetch Time
✓ 1	17:40:14	Create database Abc	1 row(s) affected	0.0019 sec
✓ 2	17:40:24	use Abc	0 row(s) affected	0.00041 sec
✓ 3	17:40:30	CREATE USER user1 IDENTIFIED BY 'password123'	0 row(s) affected	0.0068 sec

2 Create user2 with limited access:

5		
6	•	CREATE USER user2 IDENTIFIED BY 'pass456';

3 These commands create two users with login credentials in the database system.

Part B: Granting Privileges

1 Grant SELECT and INSERT privileges on Student to user1:

2 Grant ALL privileges on Course to user2:

GRANT ALL PRIVILEGES ON Course TO user2;

3 Grant WITH GRANT OPTION to user1:

GRANT SELECT ON Student TO user1 WITH GRANT OPTION;

4 Effect: user1 can further grant SELECT permission on Student to other users.

8	•	GRANT SELECT, INSERT ON Student TO user1;
9	•	GRANT ALL PRIVILEGES ON Course TO user2;
10	•	GRANT SELECT ON Student TO user1 WITH GRANT OPTION;

	Time	Action	Response	Duration / Fetch Time
✓ 1	18:10:49	GRANT ALL PRIVILEGES ON Course TO user2	0 row(s) affected	0.0013 sec

Part C: Revoking Privileges

1 Revoke INSERT permission from user1 on Student:

REVOKE INSERT ON Student FROM user1;

2 Revoke all privileges from user2 on Course:

REVOKE ALL PRIVILEGES ON Course FROM user2;

3 These commands remove previously granted permissions from the users.

12	•	REVOKE INSERT ON Student FROM user1;
13	•	REVOKE insert ON Course FROM user2;

Part D: User Deletion

- 1 Delete user2 from the database:
- 2 After deletion, login using user2 credentials will fail as the user no longer exists.

```
15 DROP USER user2;  
16
```

Part E: Additional Commands

- 1 Display all users (MySQL example):
- 2 Show privileges assigned to user1:

```
18 SELECT user FROM mysql.user;  
19 SHOW GRANTS FOR 'user1';  
20
```

	user	
	user1	
	mysql.infoschema	
	mysql.session	
	mysql.sys	
	root	